Continuous assessment BPMN

Company: Wilvo Group

Business process: Robot welding process

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FOREWORD

I would like to express my gratitude to Willem-Jan Rijken and Ralph Burgmans, who lead the welding department, including the robot welding section.

Their openness and willingness to explain and tell about our processes were not just incredibly helpful. Their ability to break down complex technical concepts into understandable terms significantly enhanced my understanding and appreciation of the intricacies involved in our production processes.

Furthermore, I was particularly impressed by their genuine curiosity regarding the purpose of my assignment. Their keen interest in understanding how the insights gained could potentially benefits to the company.

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INTRODUCTION

In this report I will provide a small introduction about the Wilvo Group and their missions and goals also this report provides a description and analysis of the various stages of the robot product workflow within the Wilvo Group organization. Special focus is placed on tasks involving with the welding robots. Where possible and necessary, suggestions and recommendations for adjustments are offered.

1 INTRODUCTION COMPANY AND CONTACT PERSON

1.1 Wilvo Group mission and goals. [1]

Wilvo Group is a company with a long history in production, specializing in metalworking, advanced digitalization of the company's infrastructure, automation, and sustainability. Their focus extends to offering enhanced capabilities and added value for clients, especially following their acquisition of GFM, a local company in CNC machining. This acquisition has enabled Wilvo Group to excel further in providing complete assemblies for large OEMs both nationally and internationally.

GFM isn't the only company that Wilvo Group has bought along the way, they've got five other businesses under their belt and have just added one more to the group. When Wilvo Group buys a company, they go all in to own it completely. That way, they can get full control and can shake things up to ensure the new addition works smoothly with Wilvo and their other businesses.

Wilvo is known for its diverse client base, competitive pricing, and their ability to tackle both complex and straightforward challenges. The company is ambitious and is striving to maintain a 'Champions League' level in its excellent quality of products and services.

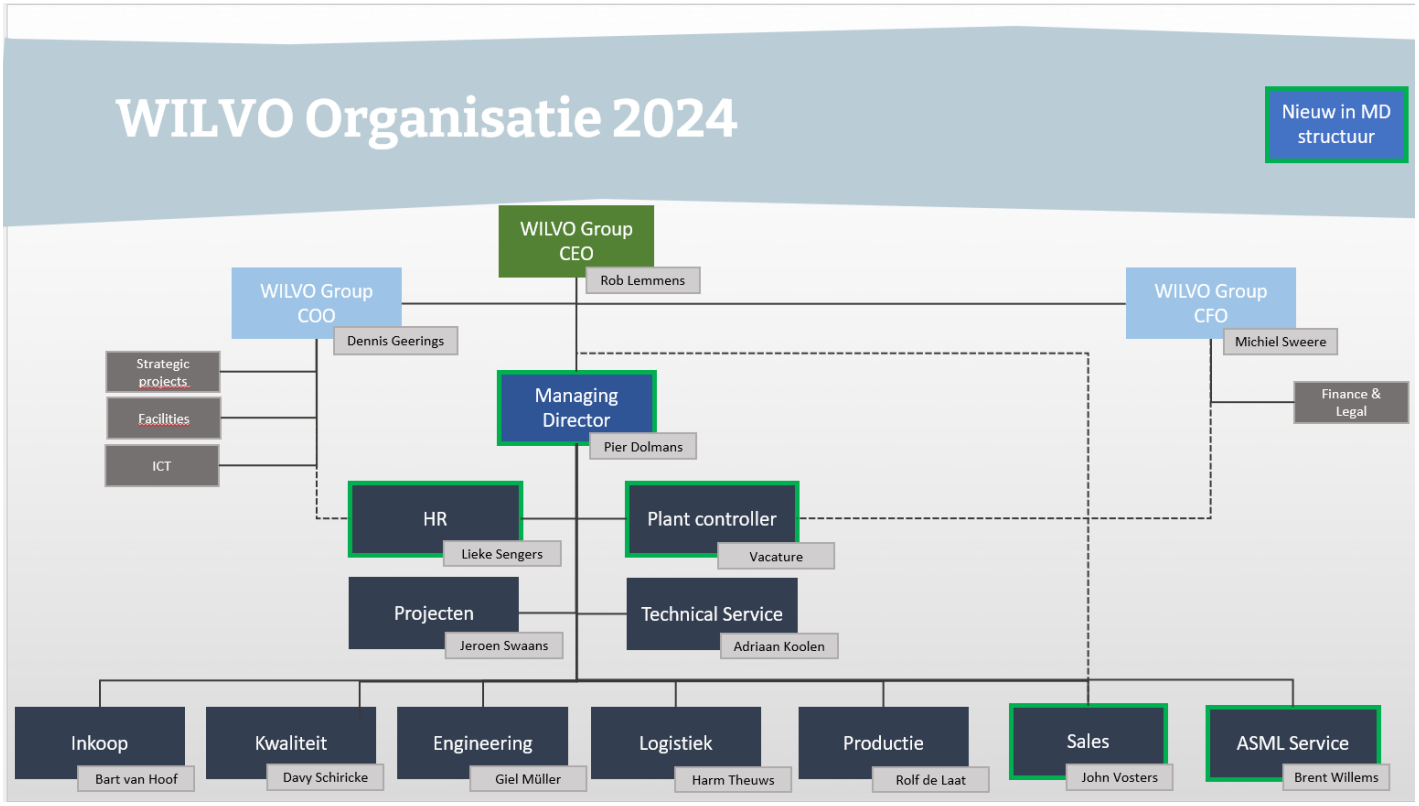
ASML is one of their largest customers. Not only does Wilvo manufacture components for their machines, but we also produce parts essential for the transportation of these machines by boat or airplane. The group does so much business with ASML that they even have their own department in the Wilvo structure. Additionally, Wilvo is the exclusive supplier to ASML for the transportation components of their machinery.

The company's expertise includes a deep involvement with sheet metal, frames, and assemblies, catering to a wide range of client requests from small, one-off projects to complex, long-running prototypes and series. Their approach is client-centric, adjusting their focus based on individual client needs. Wilvo's motto reflects their dedication: "Think, make, and organize" - emphasizing their comprehensive approach to each project, regardless of its complexity or scope.

1* Parts of this text where provided by the PR team of Wilvo Group

1.2 Wilvo Group Organizational Structure

At the top level, the organization is led by the Group CEO, COO, and CFO, who oversee the strategic direction and financial health of the company.



furthermore, the chart identifies key managers for various departments:

- The Managing Director, responsible for the overall operational management.
- HR manager, who oversees human resources and staff-related functions.
- A Plant Controller is someone who oversees the manufacturing plant's financial management. This is necessary because the Wilvo Group existed from multiple company's

Additionally, the chart lists managers of specific departments such as Projects, Technical Service, Purchasing, Quality, Engineering, Logistics, Production, and Sales. Each department is crucial for the operational flow of the company, with the production department being particularly central.

Within the Production department, it's noted that specific individuals like Willem-Jan and Ralph, who are workshop managers and there are many more, play key roles, These people? are responsible for the operational efficiency, maintenance, and if possible innovation within the manufacturing processes, ensuring that the production output meets the required standards and volumes.

The roles marked with an annotation are the newest positions within the organization, having been introduced only since this year. As you can see, there is still one function that remains open.

2 DESCRIPTION OF THE BUSINESS PROCESS

The business process of making robot products goes as follows:

A customer approaches an account manager in the sales department with a request to manufacture a product. The customer discusses the purpose of the product with the account manager. The account manager also makes other arrangements with the customer, such as the price, any post-processing of the material, and the delivery date. There are various payment terms, but these depend on the relationship the company has with the customer. Within Wilvo, they assess whether the product can be made on the robot; if not, it is made manually.

Products suitable for robotic welding are often part of large series and are usually less complex than manually welded products. Once the customer and the account manager agree on the contract, the production process begins.

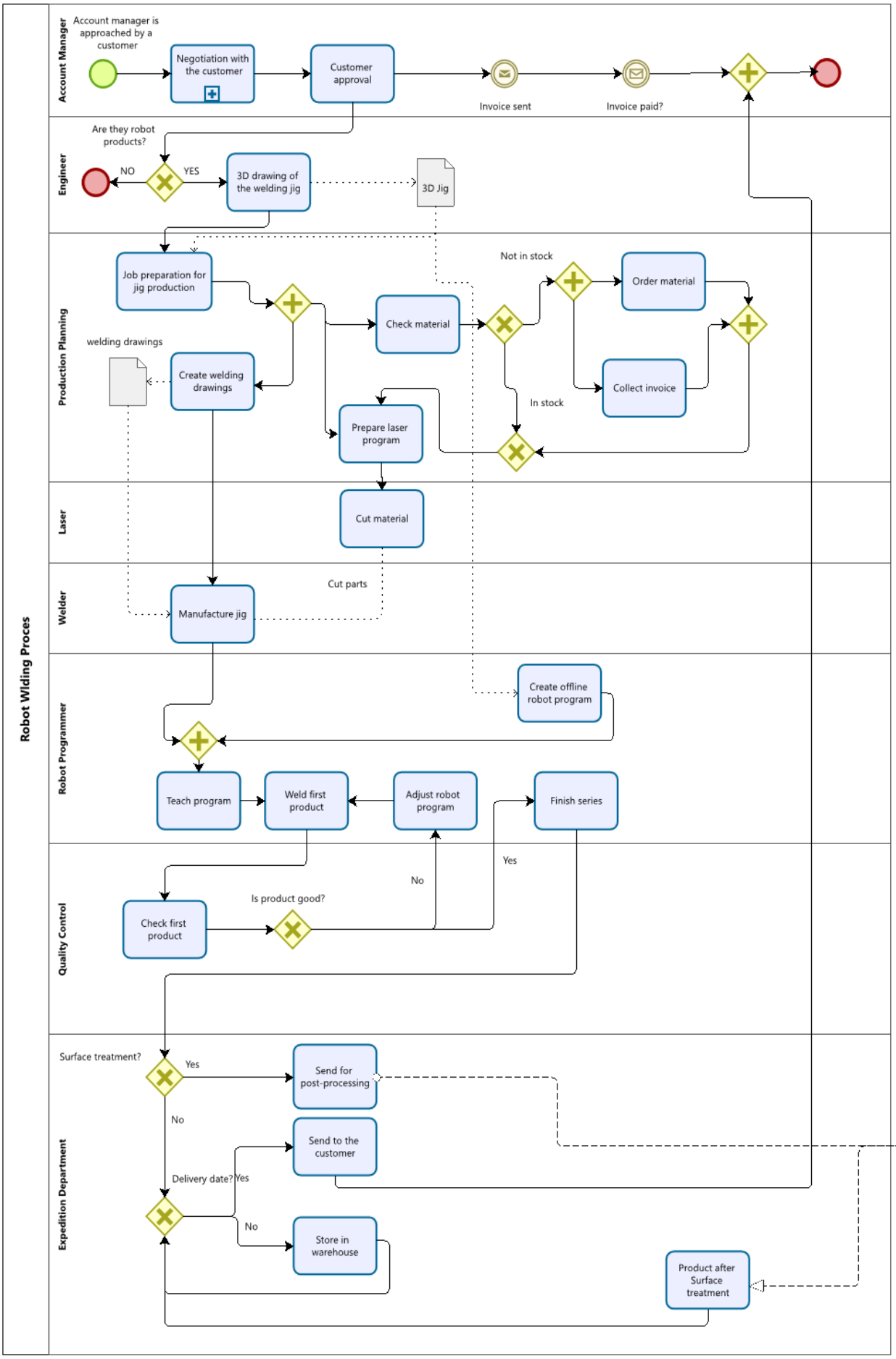
The engineer has designed the jig and passes the drawings to the production planning department. The production planning ensures that there is a laser program and that material is available to cut the jig. It also takes care of the required welding drawings. Once the laser cutting is done, the cut pieces go to a welder.

The welder assembles the jig and passes it on to the robot programmer. The programmer first creates an 'offline program' using the engineer's 3D drawing. The programmer then 'teaches' the 'offline program' to the actual robot. This is necessary because the computer simulation never matches the actual setup of the robot and jig one hundred percent. After the first product is welded, it goes to quality control department. Once approved, the series is completed and finished.

Then the products go to the expedition department. Here they decide whether to ship the products immediately or store them in the warehouse to be shipped on the agreed date. Sometimes the products are first sent for surface treatment and then sent to the customer but this depends on what the customer and account manager decided. Once the products are shipped and the invoice is paid, the process is considered completed.

3 MODELLED BUSINESS PROCESS

3.1 Robot welding process



4 SUGGESTIONS FOR IMPROVING THE BUSINESS PROCESS

The process is already quite worked out and simplified where possible. The challenging aspect is that the separate departments like quality control and the expedition department can be simplified, but its a bit difficult because of their unique internal processes, which are interlinked with various systems and components.

The engineer could check if there is enough stock of the materials when designing the jig, which would mean one less task for the production planning.

The production planning could also create the welding program for the robot, but this might lead to complications. This is because it's quite hard for the robot programmer to 'teach' the robot if they don't know how the program will move. Of course, this is solvable by allowing the robot programmer to observe and take notes while reviewing the 3D computer program made by the production planning, so the programmer isn't surprised by the robot's movements.

It's also possible to have two separate production planners who handle the subtasks: one for only the drawings and another for the laser programs, who can also handle the robot programs.

The robot programmer could also make the jig, eliminating the need for a welder. Of course, a welder knows what's important with a jig, but it's not the hardest job. Teaching the robot programmer the important aspects of making the jig would enable them to do it independently.

CONCLUSION

Reflecting on this assignment, it provided valuable insights into the concepts of "business process" and "process modelling." Although this wasn't the most complicated or IT-related process, experiencing and observing it in real life clarified many aspects. It highlighted the importance of practical application in understanding business processes and their modeling. The assignment underscored the necessity of hands-on experience in grasping the dynamics of process management and optimization, offering a more tangible perspective on theoretical principles.